Yeo Meng Han

📞+65-98150541 | ✉️yeo_menghan@u.nus.edu | 🔗linkedin.com/in/yeo-meng-han | 🌐yeo-menghan.com/

Applied AI Engineer, based in Singapore, building and deploying production AI systems for computer vision, speech, and multimodal applications across cloud, mobile, and edge

WORK / INTERNSHIP EXPERIENCE

Embedded Computer Vision Engineer, Avetics Global **Feb 26 - Present**

- Developed and optimised YOLO-based edge computer vision pipelines for autonomous drones, meeting real-time latency needs while maintaining robust aerial object tracking across resource-constrained embedded platforms.
- Accelerated edge AI inference by 60% through parallelising SAHI tiled inference on the RK3588 (Rock 5C) NPU, and optimised models for deployment using RKNN, Hailo, and Axelera quantisation toolchains.
- Built a Docker-based deployment platform enabling remote drone registration, software provisioning, and remote updates through a centralized management dashboard.

AI Engineer Intern, NUS Office of Internal Audit **Sep 25 - Feb 26**

- Engineered an end-to-end invoice fraud detection system using OCR and document similarity techniques, containerised using Docker for scalable internal use to cut manual audit time.
- Normalised noisy travel-location claims using embeddings and a local SLM, preserving data privacy while enabling reliable geocoding and compliance checks.

Machine Learning Engineer Intern, Changi Airport Group **May 25 - Aug 25**

- Improved taxi demand forecasting accuracy from 48% to 62% using XGBoost and time series decomposition, shipped to production on AWS with automated retraining.
- Analysed 18,000+ flight turnarounds to identify operational bottlenecks and reduce departure delays.

Research Assistant, Cisco-NUS Accelerated Digital Economy Corporate Laboratory **Jan 25 - May 25**

- Developed a real-time overhead multi-person tracking system using YOLOv8 and DeepSORT, achieving 15 FPS on GPU-enabled servers with UDP streaming to support live digital twin analytics.

AI & Computer Vision Engineer Intern, Rockship (NUS Overseas College Vietnam) **Jan 24 - Jun 24**

- Boosted YOLOv8s object detection model accuracy by 20% through targeted labeling and deployed the model in an ARKit iOS app for precise nail alignment and distance measurement in construction.
- Architected a RAG n8n copilot assistant to generate simple n8n workflows, with LangSmith monitoring.

Robotics Engineer Intern, DSTA Land Systems **May 23 - Jul 23**

- Integrated real-time visual mapping and localisation (SLAM) onto a mobile robot platform for autonomous navigation in GPS-denied environments.

PROJECTS

TikTok TechJam 2025 - iOS PII Detection Project | Devpost **Aug 25**

- Led end-to-end mobile development in Swift, integrating distilBERT model via CoreML to enable on-device PII detection and blurring.

Smart Delivery Allocator (Grains) | Github **Oct 25**

- Built a GPT-4.1-powered AI agent to optimise catering order allocation under real-world scheduling and routing constraints, with deterministic validation to improve assignment reliability.
- Built a Python and Streamlit/Folium application to visualise and manage delivery routes.

EDUCATION

National University of Singapore **Jul 25 - Dec 26 (Expected)**

Master of Science, Data Science and Machine Learning

- Relevant Coursework: Scalable Distributed Computing, Deep Learning, Cloud Computing

National University of Singapore **Aug 22 - Jul 25**

Bachelor of Engineering, Major in Computer Engineering

- GPA: 4.51 / 5.00 (First Class Honours)
- NUS Engineering Scholarship Programme (awarded to top engineering students)

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C/C++, Swift (iOS)
- **ML / CV:** PyTorch, TensorFlow, scikit-learn, OpenCV, Pandas, NumPy
- **Generative AI:** Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, Diffusion Models
- **MLOps & Deployment:** Docker, FastAPI, CI/CD (GitHub Actions, Bitbucket Pipelines), Model Monitoring (LangSmith, LangFuse), Model Optimisation (ONNX, CoreML)
- **Cloud**: AWS, Google Cloud Platform - Professional ML Engineer (GCP)